

Summary

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0.1 Extreme Value Theorem

A function f has a **global maximum** at $x = a$, if $f(a) \geq f(x)$, for all x in the domain of f .

A function f has a **global minimum** at $x = a$, if $f(a) \leq f(x)$, for all x in the domain of f .

Theorem 1 (Extreme Value Theorem (EVT)). If a function f is **continuous on the closed interval** $[a, b]$, then f attains both a global maximum and a global minimum on the closed interval $[a, b]$.

Note: If a function f has a global extremum (minimum or maximum) at $x = c$, then c is either a boundary point (which means that $c = a$ or $c = b$), or f has a critical point at $x = c$.

So, if we have to find an extreme value of f on $[a, b]$, we should check all the boundary and all the critical points of f and compare the values of f at those points. The biggest of those values is the maximum value of f on $[a, b]$, and the smallest one is the minimum value of f on $[a, b]$.

0.2 The Mean Value Theorem

Theorem 2 (The Mean Value Theorem (MVT)). If a function f is **continuous on the closed interval** $[a, b]$, and **differentiable on** (a, b) , then there exists a point c in the open interval (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Note 1a: The value $\frac{f(b) - f(a)}{b - a}$ is the slope of the (secant) line through the points $(a, f(a))$ and $(b, f(b))$.

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Note 1b: The line tangent to the graph of f at the point where $x = c$ is **parallel** to the line through the points $(a, f(a))$ and $(b, f(b))$.

Note 2a: The value $\frac{f(b) - f(a)}{b - a}$ is the average rate of change of f on the interval $[a, b]$.

Note 2b: The instantaneous rate of change f at the point where $x = c$ is equal to the average rate of change of f on $[a, b]$.